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Review

# Mitochondria in health and disease

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ABSTRACT

Mitochondrial biology has become an area of intense research owing to the unique physiology of the organelle and its role in several types of cancers and other disorders. It has been found that mitochondria-encoded proteins, mitochondrial DNA and even RNA influence the functioning of the cell in more ways than were previously imagined. This may contribute to disease phenotypes which require detailed investigation and communication to the community health care providers. Additionally, this provides several novel avenues in drug designing against various cancers, neurodegenerative diseases and other metabolic disorders. The sixth annual conference of the Society for Mitochondrial Research and Medicine – India (SMRM) titled, 'Mitochondria in Health and Disease' was organized by Rana P. Singh at the School of Life Sciences, Jawaharlal Nehru University in New Delhi, India from 10th to 11th February 2017. The underlying objective of the conference was to provide a platform to discuss the recent advances in basic and translational research in mitochondrial biology and diseases. The conference aimed to translate academic research into clinical practice by providing a forum for basic researchers and clinicians to share their knowledge and build collaborations towards development of advanced therapeutic in mitochondrial diseases. To facilitate the knowledge-sharing, six major themes for the scientific sessions were (1) understanding of mitochondrial biology in disease progression, (2) advances in basic and translational mitochondrial research, (3) mitochondria in evolution and development, (4) targeting mitochondria for cancer prevention and treatment, (5) mitochondria in metabolic and neurological disorders and (6) mitochondria in stem cell and regeneration biology. This report summarizes the major outcomes of the discussions at the conference.

## 1. Novel agents targeting mitochondria

In recent times, several compounds that target mitochondria have been discovered for the treatment of various diseases such as cancer and metabolic disorders (He et al., 2015; Peng et al., 2017). Since mitochondria are the powerhouse of the cell, agents that target them by disrupting their membrane potential or DNA are cytotoxic and thus, effective anticancer drugs (Peng et al., 2017).

The first session of the conference was focused on novel chemotherapeutic agents targeting mitochondria and its mechanism of action in treatment of cancer. Rajesh Agarwal from Skaggs School of Pharmacy, University of Colorado Denver, USA, presented the dual efficacy of Silibinin, a natural agent, in targeting colorectal cancer cells (Dheeraj et al., 2017). For the first time, his team has proven that Silibinin activates both caspase-dependent apoptotic and caspase-independent autophagic cell death in these cells and could be used as an adjunct in anticancer treatment (Raina et al., 2013).

Sanjay Gupta from Case Western Reserve University, University Hospitals Cleveland Medical Center, Cleveland, USA presented the results of a follow-up study on the use of simvastatin-metformin (SIM + MET) drug combination to bioenergetically reprogram and thus, kill cancer cells (Babcook et al., 2014). His group has shown that simvastatin-metformin (SIM + MET) drug combination can subvert resistance to chemotherapy acquired by cancer cells due to the Warburg effect (Babcook et al., 2014). This was tested on CRPC C4-2B apoptosis-resistance cancer cells. Treating these cells with SIM + MET in 1:500

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ratio induced necroptosis, which was confirmed by observing increased Ripk1 and Ripk3 protein expression and increased necrosome formation (Babcook et al., 2014). Autophagy was ruled out as the operating cell death mechanism since chloroquine was unable to unable to rescue cell viability.

Sudin Bhattacharya from Department of Cancer Chemoprevention, Chittaranjan National Cancer Institute, India shared his work on small molecules as adjuvants to chemotherapy (Hajra et al., 2018). His group has identified one such compound named prenylated benzylidine-thiazolidine-2,4-dione (PBT) which showed promising anti-tumorigenic activity. When administered in combination with cyclophosphamide to Swiss albino mice with breast adenocarcinoma cells, PBT resulted in increased tumor cell death, further confirmed by DNA damage assay. PBT was found to induce the mitochondrial apoptotic pathway via down-regulation Bcl-2 and up-regulation of Bax. Additionally, PBT has shown to inhibit angiogenesis, which was confirmed by decreased levels of the pro-angiogenic mediators such as VEGF-A and MMP-9. Since PBT acts as a sensitizer in cyclophosphamide-based chemotherapy, it may also lead to a decrease in side effects such as hepatotoxicity and genotoxicity to normal cells.

## 2. Altered mitochondrial function in disease progression

Mitochondrial dysfunction can lead to a wide variety of disorders and diseases (Govindaraj et al., 2014; Kabekkodu et al., 2014; Roshan et al., 2012; Singh and Modica-Napolitano, 2017; Singh et al., 2014). These are generally due to a lack of energy production in the affected cells. Thus, such dysfunctions generally present themselves in the form of skeletal muscle or neuromuscular disorders. Mitochondrial dysfunction can also lead to cancer (Kabekkodu et al., 2014; Singh and Modica-Napolitano, 2017). Some other common mitochondrial disorders include MELAS, LHON and MIRAS (Khan et al., 2017; Vandana et al., 2016).

Lene Juel Rasmussen from Center for Healthy Aging, Department of Cellular and Molecular Medicine, University of Copenhagen, Denmark elucidated the importance of the Rev1 protein in aging and age-related disorders (Fakouri et al., 2017). Her group has extensively studied the function of Rev1 protein and its role in mutagenic translesion synthesis (TLS) of DNA, which is carried out by translesion synthesis polymerases. Rev1 deficient mice were used as a model for this study and they were shown to exhibit mild progeria-like symptoms. The molecular mechanisms causing progeria in these mice were further investigated and it was found that the absence of Rev1 leads to mitochondrial dysfunction and abnormal morphology, triggering DNA damage sensor PARP-1, $NAD^+$ depletion and subsequent cellular stress (Fakouri et al., 2017).

P. Sundaresan from Department of Molecular Genetics, Aravind Medical Research Foundation, Madurai, India elaborated on the results of mitochondrial gene expression signature (Mitoscriptome) in diabetic retinopathy (Burdon et al., 2015). His group analyzed the gene expression levels of more than 1000 genes, which included 13 mtDNA-encoded genes and nuclear encoded mitochondrial genes in diabetic retinopathy. Gene expression analysis of human cadaver eye with diabetic retinopathy to age matched controls identified 59 differentially expressed genes which have known mitochondrial function. Among them, 8 of these genes were upregulated and 51 were downregulated. This data can serve as a reference in further studied and diagnosis of diabetic retinopathy.

K. Thangaraj from CSIR-Centre for Cellular and Molecular Biology (CCMB), India discussed the clinical and genetic heterogeneity of mitochondrial diseases in the diverse Indian population. His group has shown that mtDNA mutations usually present themselves as disorders of energy-requiring organ systems, such as the neuromuscular system and identified several mitochondrial and nuclear mutation in patients with neuromuscular dysfunction with relevance to different population types in India (Govindaraj et al., 2014; Khan et al., 2017; Paramasivam et al., 2016; Sonam et al., 2017; Vandana et al., 2016; Vanniarajan et al., 2011).

Gayathri N. from Department of Neuropathology, NIMHANS, India discussed various pathological conditions in skeletal muscle caused by dysfunctions of the mitochondrial respiratory chain (Debashree et al., 2018). She elaborated on her group's attempts to analyze the genetic basis of chronic progressive external ophthalmoplegia (CPEO) in the Indian context.

Anshu Bharadwaj from CSIR-Institute of Microbial Technology, India talked about the development of the first fingerprinting algorithm (FROG) to barcode genome variation effect on phenotype due to the complexity to establish genotype-phenotype correlation (Abinaya et al., 2015). FROG can tag variation data based on its location, function and interactions and assign bit scores to each of these properties and generate a fingerprint based on the combination of these properties. It uses six blocks – Chromosome, DNA, RNA, Protein, variations, and functional interactions to describe various annotations. Her group is also working on designing a web-based platform MitoLINK to facilitate whole mitochondrial genome analysis for multiple diseases.

Suhel Parvez from Department of Toxicology, Jamia Hamdard, India talked about his research on mitochondrial ATP sensitive potassium channel blocker 5-hydroxydecanoate (5- HD) and Calcium alter the permeability transition pore (PTP) in isolated brain mitochondria pretreated with melatonin (Mel) and cyclosporine A (CsA) (Waseem et al., 2016). This was tested by evaluating mitochondrial membrane potential, measuring ROS and mitochondrial respiration and mitochondrial swelling, the latter being stimulated by exposure of Calcium ions and 5-HD associated by mPTP opening as depicted by modulation of CsA and Mel. They concluded that inhibition of mPT is a probable mechanism of CsA's neuroprotective actions and can therefore be used in the future for treatment of neurodegenerative diseases.

Arttatrana Pal from School of Life Sciences, Mahatma Gandhi Central University, India have presented his work on IRF-3 signaling via mTOR pathway in retinal neuronal cell model. His group has shown that restoring IRF-3 signaling via mTOR pathway presents and effective therapeutic modality to reduce neuronal complications in diabetic retinopathy (Samanta et al., 2014).

## 3. Mitochondria in cancer prevention and therapeutics

As an intrinsic part of the cell, mitochondria play a pivotal role in cancer progression (Choudhury and Singh, 2017; Kabekkodu et al., 2014; Singh and Modica-Napolitano, 2017). Studying the crosstalk between the mitochondrial and nuclear genome along with the molecular functioning of mitochondria during cancer can help identify therapeutic agents that can target mitochondria to prevent and/or manage cancer (Carden et al., 2017).

K. Satyamoorthy from School of Life Sciences, Manipal University, India presented data on the crosstalk between nuclear encoded and mitochondria encoded microRNA which might play a significant role in facilitating cancer progression(Unpublished data). This hypothesis was tested in cervical cancer model by performing small RNA sequencing of normal cervical epithelium, cervical intraepithelial neoplasia and squamous cell carcinoma, as well as of normal healthy control. His group identified 7 significant nuclear encoded miRNAs that showed differential expression in normal healthy control cells versus normal cervical epithelium cells. They have also identified novel mitochondria encoded miRNAs and their potential cellular targets. This data showed that nuclear and mitochondrial miRNA can act in tandem to alter the tumor microenvironment at the cellular level.

Erich Gnaiger from Medical University Innsbruck, Austria discussed his research on the role of defective mitochondrial energy metabolism in malignant diseases (Gnaiger, 2003; Schopf et al., 2016). His group has extensively studied the defects in oxidative phosphorylation (OXPHOS) by high-resolution respirometry (HRR) in benign and malignant tissues and showed reprogramming of mitochondrial pathways and shift of substrate utilization preference in the malignant tissues which had a higher mutation load (based on whole mtDNA sequencing). He has also explained that the OXPHOS capacity of the tumor cells was completely restored when pyruvate was used as a substrate instead of glutamate and succinate.

Sujit Bhutia from National Institute of Technology, Rourkela, India gave an insight into the crosstalk between autophagy and apoptosis via the Ulk1 protein (Saha et al., 2018). His group produced Ulk1 overexpressing-HeLa cell line and studied the cells' apoptosis-inducing potential in the presence of cisplatin. These cells showed increased hallmarks of apoptosis including increased percentage of annexin V, increased reactive oxygen species and increased activity of caspases 9 and 3, among other hallmarks. The molecular mechanism was investigated and it was found that Ulk1 triggers autophagy (particularly mitophagy), leading to a fall in ATP, overworking of the ETC of the non-mitophagic mitochondria, causing increased ROS production and subsequent apoptotic cell death. Ulk1 also increases the mitochondrial superoxide level by entering into the mitochondria and inhibiting manganese dismutase (MnSOD) activity.

Abid Hamid Dar from CSIR-Indian Institute of Integrative Medicine, India presented his work on novel P13K inhibitor – 3-Dihydro-2-(naphthalene-1-yl) quinazolin-4(1H)-one (DHNQ) which would inhibit the P13K enzyme activity at transcriptional and translational levels in colorectal cancer(CRC), as P13K pathway signaling up-regulates cancer cell proliferation and metastasis (Hussain et al., 2016a,b). DHNQ inhibited the Warburg effect, decreased the expression of cyclins responsible for G1 arrest, decreased Bcl2/Bax ratio after mitochondrial membrane potential loss, ROS generation and an increase in cytosolic $Ca^{+2}$ load - all of which is responsible for decreased CRC cell proliferation and survival.

## 4. Mitochondrial genome, epigenetics and cancer

The mitochondrial DNA, its mutations and its epigenetic regulation has become an area of intense research in current times (Carden et al., 2017; Minocherhomji et al., 2012). Mitochondria may provide a new route that can be exploited to selectively kill cancer cells or slow down their growth.

R.N.K. Bamezai from School of Life Sciences, Jawaharlal Nehru University, India presented his work on genetic and epigenetic modulation of the mitochondrial DNA and its role in cancer progression (Saini et al., 2018). Using a codon optimized approach, his group identified mutations in the mitochondrial gene MT-ND3 and MT-ND5 genes that lead to pro-cancerous phenotypes. His group also identified *DNMT1* isoform 3 as the enzyme that epigenetically regulated mitochondrial gene expression. Ectopic overexpression of this enzyme was found to result in its localization to the mitochondria and subsequent changes in mitochondrial biology and function.

## 5. Recent advances in mitochondrial biology

Since mitochondrial biology has become a new research focus area, there was a session on recent advances in on this subject. Mitochondrial dynamics is an area that is still relatively unexplored and there lies an immense potential to design therapeutic agents for multi-system diseases and disorders by understanding and manipulating mitochondrial biology.

Keshav K. Singh from Center for Aging and UAB comprehensive Cancer Center, University of Alabama at Birmingham, USA reported the discovery of increased copies of nuclear mtDNA (NUMT) in colorectal adenocarcinomas (Zhang and Singh, 2014). This indicates a link between mtDNA and genomic instability in the nucleus, thus naming the phenomenon numtogenesis. His group has identified that on an average, up to 4.2-fold more somatic NUMTs were present in colorectal adenocarcinoma than in matched normal genomes. Increases numtogenesis was observed with higher mortality. Identified YME1L1, a human homolog of yeast YME1 which is mutated in colorectal tumor and that inactivation of YME1L1 causes increased transfer of mtDNA in the nuclear genome and it is one of the first NUMT suppressor gene. Study also reveals sex based differences in frequency of NUMT occurrence and that NUMT occurrence in tumours suggests that it can be used as a biomarker for tumorigenesis.

Robert Pitceathly from MRC Center for Neuromuscular Diseases, UCL Institute of Neurology UK gave an overview of recent mitochondrial disease-related discoveries at the UCL Institute of Neurology. He explained how these discoveries are being translated into clinical practice to help diagnose and manage a wide variety of mitochondrial diseases (Bugiardini et al., 2017; Pitceathly and Viscomi, 2016).

Anil Shanker from Translational and Clinical Immunology, Vanderbilt University, USA explained his finding that cross-regulation between the $CD8^+$ T cells and NK cells can prevent the development of antigen-escape tumor variants (Pellom Jr. et al., 2017). His group used an engineered nanofiber matrix to provide a controlled 3D interaction environment to the lymphocytes, and found that the $CD8^+$ T cells formed intercellular contacts with naïve NK cells but the naïve $CD8^+$ T cells did not. The cross-regulation between the cells disappeared in system with no physical cell to cell interaction. The physical interaction between the cells regulates mitochondrial $Ca^{2+}$ uptake in activated NK and $CD8^+$ T cells. He concluded that mitochondrial $Ca^{2+}$ dynamics as a key biological controller for cellular remodeling.

Naresh Babu V Sepuri's group from University of Hyderabad showed that for proper mitochondrial function, redox regulation of Grpe (co-chaperone of Hsp70) facilitated by methionine sulfoxide reductase (Mxr1/MsrA & Mxr2/MsrB) is essential. His group has shown that Mge1 (yeast homolog of bacterial GrpE) acts as an oxidative sensor to regulate Hsp70 in mitochondria. Deletion of mitochondrial Mxr2 makes yeast cells more sensitive to oxidative stress than Mxr1 in cytosol (Karri et al., 2018; Marada et al., 2016).

Chandramani Pathak from Indian Institute of Advanced Research, Gujrat, India talked about the role of FADD and $cFLIP_L$ as modulators of mitochondria-associated apoptosis apart from their role in death receptor signaling (Ranjan and Pathak, 2016). His group found that induced expression of FADD damages mitochondrial integrity and membrane potential by altering the expression of Bcl-2 and cytochrome *c*. Ectopic expression of $cFLIP_L$ helps reverse resistance of cancer cells to apoptosis inducers. They have shown that silencing $cFLIP_L$ and inducing the expression of FADD leads to accumulation of intracellular ROS, JNK1 activation and subsequent apoptosis.

Binay Chaubey's group from University of Calcutta, India presented data on Efavirenz (EFV), an essential component of HAART (combination of different drug classes) used in HIV treatment, mediated mitochondrial dysfunction (Ganta et al., 2017). His group has shown that human hepatoma cells (Huh 7.5) treated with EFV facilitates permeabilization of mitochondrial outer membrane, change in mitochondrial membrane potential by releasing cytochrome *c* and subsequent changes in mitochondrial morphology and mitochondria mediated apoptosis.

Anil K. Mantha from Central University of Punjab presented evidence for the neuro-modulatory role of *APE1* enzyme in Alzheimer's disease (Kaur et al., 2015). They have found that mitochondrial dysfunction and oxidative stress are key factors in the progression of Alzheimer's disease. The group's preliminary findings suggest that ADO enzyme, a mammalian thiol-dioxygenase, is a key partner of *APE1* and upregulated in the brain.

## 6. Mitochondria in stem cell and regeneration

The role of mitochondria in stem cells and their regenerative capacity needs to be investigated to understand cellular energetics and functioning of stem cells used in clinics. This may help design better regenerative therapeutic strategies for the future.

Nick Jones from Imperial College of London, UK talked about mitochondrial heterogeneity and the role of stochasticity in mitochondrial physiology and genetics (Aryaman et al., 2017). He explained how changes in mitochondrial size and density within a tissue can also have physiological consequences.

Suresh Mishra from University of Manitoba, Canada explained the roles of Prohibitin (PHB) in adipose, immune functions at systemic level & cell compartment, tissue-specific functions by developing two transgenic mice (Zi Xu et al., 2018). This was done by expressing PHB and Y114f-PHB(m-PHB), a phospho mutant of PHB, from aP2 (adipocyte protein-2) respectively. Both the transgenic mice develop obesity in sex-neutral manner suggesting m-PHB retains mitochondria mediated adipogenic functions of PHB, obesity-related metabolic dysregulation in male-specific manner suggesting sex dimorphic effect of PHB in adipose tissue. It was observed that male-PHB mice develop hepatocellular carcinoma while m-PHB mice develop lymph node tumours (with aging), suggesting phosphorylation of tyrosine-114 in PHB plays a role in macrophages and dendritic cells. This reveals that both mice share metabolic phenotype but differ in immune phenotype. Development of tumours & autoimmune diabetes in a mutually exclusive context dependent manner suggesting a role of PHB in immune checkpoint pathways.

Sujata Mohanty from Stem Cell Facility, AIIMS, India presented her group's study to understand the rescue mechanism of ASC via mitochondria in an in vitro cell damage model. Mesenchymal stem cells from bone marrow, adipose tissue and dental pulp were evaluated for their rescue potential in the presence of damaged human fibroblast cells. Rescue potential was evaluated in terms of mitochondrial morphology and membrane potential, ATP synthesis and rate and degree of mitochondrial transfer. Additionally, the group also looked to see whether there were any mitochondrial changes occurring in mesenchymal stem cells after they differentiate into cells of cardiac and neuronal lineage.

Anurag Aggarwal from Institute of Genomics and Integrative Biology, India described the role of mesenchymal stem cells (MSC) as a source of mitochondria donors to stressed or dying lung epithelial cells (Ahmad et al., 2014). Intercellular nanotubes (as the highway) & *Miro1*, along with associated proteins (as wheels) aid in transfer of mitochondrial cargo. Aging and metabolic stress lead to impaired donor capacity in MSC. MSC, which show high efficacy in lung injury & asthma models, were bioengineered to enhance their efficiency.

On the second day of the conference, there was also one session for the Prof. A.R. Rao Young Scientist Award with four presentations on different diseases and the role of mitochondria in these diseases. There was also one session for the Prof. A.R. Rao Young Researcher Award with another four presentations about mitochondrial biology. Lastly, there were several poster presentations on a wide variety of topics from mitochondrial signaling to mitochondrial microRNA and other related topics.

## References


Abinaya, E., Narang, P., Bhardwaj, A., 2015. FROG - fingerprinting genomic variation ontology. PLoS One 10, e0134693.

Ahmad, T., Mukherjee, S., Pattnaik, B., Kumar, M., Singh, S., Kumar, M., Rehman, R., Tiwari, B.K., Jha, K.A., Barhanpurkar, A.P., Wani, M.R., Roy, S.S., Mabalirajan, U., Ghosh, B., Agrawal, A., 2014. Miro1 regulates intercellular mitochondrial transport & enhances mesenchymal stem cell rescue efficacy. EMBO J. 33, 994–1010.

Aryaman, J., Hoitzing, H., Burgstaller, J.P., Johnston, I.G., Jones, N.S., 2017. Mitochondrial heterogeneity, metabolic scaling and cell death. BioEssays 39.

Babcook, M.A., Shukla, S., Fu, P., Vazquez, E.J., Puchowicz, M.A., Molter, J.P., Oak, C.Z., Maclennan, G.T., Flask, C.A., Lindner, D.J., Parker, Y., Daneshgari, F., Gupta, S., 2014. Synergistic simvastatin and metformin combination chemotherapy for osseous metastatic castration-resistant prostate cancer. Mol. Cancer Ther. 13, 2288–2302.

Bugiardini, E., Poole, O.V., Manole, A., Pittman, A.M., Horga, A., Hargreaves, I., Woodward, C.E., Sweeney, M.G., Holton, J.L., Taanman, J.W., Plant, G.T., Poulton, J., Zeviani, M., Ghezzi, D., Taylor, J., Smith, C., Fratter, C., Kanikannan, M.A., Paramasivam, A., Thangaraj, K., Spinazzola, A., Holt, I.J., Houlden, H., Hanna, M.G., Pitceathly, R.D.S., 2017. Clinicopathologic and molecular spectrum of RNASEH1-related mitochondrial disease. Neurol. Genet. 3, e149.

Burdon, K.P., Fogarty, R.D., Shen, W., Abhary, S., Kaidonis, G., Appukuttan, B., Hewitt, A.W., Sharma, S., Daniell, M., Essex, R.W., Chang, J.H., Klebe, S., Lake, S.R., Pal, B., Jenkins, A., Govindarjan, G., Sundaresan, P., Lamoureux, E.L., Ramasamy, K., Pefkianaki, M., Hykin, P.G., Petrovsky, N., Brown, M.A., Gillies, M.C., Craig, J.E., 2015. Genome-wide association study for sight-threatening diabetic retinopathy reveals association with genetic variation near the GRB2 gene. Diabetologia 58, 2288–2297.

Carden, T., Singh, B., Mooga, V., Bajpai, P., Singh, K.K., 2017. Epigenetic modification of miR-663 controls mitochondria-to-nucleus retrograde signaling and tumor progression. J. Biol. Chem. 292, 20694–20706.

Choudhury, A.R., Singh, K.K., 2017. Mitochondrial determinants of cancer health disparities. Semin. Cancer Biol. 47, 125–146.

Debashree, B., Kumar, M., Keshava Prasad, T.S., Natarajan, A., Christopher, R., Nalini, A., Bindu, P.S., Gayathri, N., Srinivas Bharath, M.M., 2018. Mitochondrial dysfunction in human skeletal muscle biopsies of lipid storage disorder. J. Neurochem. 145, 323–341.

Dheeraj, A., Rigby, C.M., O'Bryant, C.L., Agarwal, C., Singh, R.P., Deep, G., Agarwal, R., 2017. Silibinin treatment inhibits the growth of hedgehog inhibitor-resistant basal cell carcinoma cells via targeting EGFR-MAPK-Akt and hedgehog signaling. Photochem. Photobiol. 93, 999–1007.

Fakouri, N.B., Durhuus, J.A., Regnell, C.E., Angleys, M., Desler, C., Hasan-Olive, M.M., Martin-Pardillos, A., Tsaalbi-Shtylik, A., Thomsen, K., Lauritzen, M., Bohr, V.A., de Wind, N., Bergersen, L.H., Rasmussen, L.J., 2017. Rev1 contributes to proper mitochondrial function via the PARP-NAD(+)-SIRT1-PGC1alpha axis. Sci. Rep. 7, 12480.

Ganta, K.K., Mandal, A., Chaubey, B., 2017. Depolarization of mitochondrial membrane potential is the initial event in non-nucleoside reverse transcriptase inhibitor efavirenz induced cytotoxicity. Cell Biol. Toxicol. 33, 69–82.

Gnaiger, E., 2003. Oxygen conformance of cellular respiration. A perspective of mitochondrial physiology. Adv. Exp. Med. Biol. 543, 39–55.

Govindaraj, P., Khan, N.A., Rani, B., Rani, D.S., Selvaraj, P., Jyothi, V., Bahl, A., Narasimhan, C., Rakshak, D., Premkumar, K., Khullar, M., Thangaraj, K., 2014. Mitochondrial DNA variations associated with hypertrophic cardiomyopathy. Mitochondrion 16, 65–72.

Hajra, S., Patra, A.R., Basu, A., Bhattacharya, S., 2018. Prevention of doxorubicin (DOX)-induced genotoxicity and cardiotoxicity: effect of plant derived small molecule indole-3-carbinol (I3C) on oxidative stress and inflammation. Biomed Pharmacother 101, 228–243.

He, H., Li, D.W., Yang, L.Y., Fu, L., Zhu, X.J., Wong, W.K., Jiang, F.L., Liu, Y., 2015. A novel bifunctional mitochondria-targeted anticancer agent with high selectivity for cancer cells. Sci. Rep. 5, 13543.

Hussain, A., Qazi, A.K., Mupparapu, N., Guru, S.K., Kumar, A., Sharma, P.R., Singh, S.K., Singh, P., Dar, M.J., Bharate, S.B., Zargar, M.A., Ahmed, Q.N., Bhushan, S., Vishwakarma, R.A., Hamid, A., 2016a. Modulation of glycolysis and lipogenesis by novel PI3K selective molecule represses tumor angiogenesis and decreases colorectal cancer growth. Cancer Lett. 374, 250–260.

Hussain, A., Qazi, A.K., Mupparapu, N., Kumar, A., Mintoo, M.J., Mahajan, G., Sharma, P.R., Singh, S.K., Bharate, S.B., Zargar, M.A., Ahmed, Q.N., Mondhe, D.M., Vishwakarma, R.A., Hamid, A., 2016b. A novel PI3K axis selective molecule exhibits potent tumor inhibition in colorectal carcinogenesis. Mol. Carcinog. 55, 2135–2155.

Kabekkodu, S.P., Bhat, S., Mascarenhas, R., Mallya, S., Bhat, M., Pandey, D., Kushtagi, P., Thangaraj, K., Gopinath, P.M., Satyamoorthy, K., 2014. Mitochondrial DNA variation analysis in cervical cancer. Mitochondrion 16, 73–82.

Karri, S., Singh, S., Paripati, A.K., Marada, A., Krishnamoorthy, T., Guruprasad, L., Balasubramanian, D., Sepuri, N.B.V., 2018. Adaptation of Mge1 to oxidative stress by local unfolding and altered interaction with mitochondrial Hsp70 and Mxr2. Mitochondrion. http://dx.doi.org/10.1016/j.mito.2018.04.003. (2018 Apr 9, pii: S1567-7249(17)30346-X).

Kaur, N., Dhiman, M., Perez-Polo, J.R., Mantha, A.K., 2015. Ginkgolide B revamps neuroprotective role of apurinic/apyrimidinic endonuclease 1 and mitochondrial oxidative phosphorylation against Abeta25-35 -induced neurotoxicity in human neuroblastoma cells. J. Neurosci. Res. 93, 938–947.

Khan, N.A., Govindaraj, P., Soumittra, N., Sharma, S., Srilekha, S., Ambika, S., Vanniarajan, A., Meena, A.K., Uppin, M.S., Sundaram, C., Bindu, P.S., Gayathri, N., Taly, A.B., Thangaraj, K., 2017. Leber's hereditary optic neuropathy-specific mutation m.11778G > A exists on diverse mitochondrial haplogroups in India. Invest. Ophthalmol. Vis. Sci. 58, 3923–3930.

Marada, A., Karri, S., Singh, S., Allu, P.K., Boggula, Y., Krishnamoorthy, T., Guruprasad, L., Sepuri, N.B.V., 2016. A single point mutation in mitochondrial Hsp70 cochaperone Mge1 gains thermal stability and resistance. Biochemistry 55, 7065–7072.

Minocherhomji, S., Tollefsbol, T.O., Singh, K.K., 2012. Mitochondrial regulation of epigenetics and its role in human diseases. Epigenetics 7, 326–334.

Paramasivam, A., Meena, A.K., Pedaparthi, L., Jyothi, V., Uppin, M.S., Jabeen, S.A., Sundaram, C., Thangaraj, K., 2016. Novel mutation in C10orf2 associated with multiple mtDNA deletions, chronic progressive external ophthalmoplegia and premature aging. Mitochondrion 26, 81–85.

Pellom Jr., S.T., Dudimah, D.F., Thounaojam, M.C., Uzhachenko, R.V., Singhal, A., Richmond, A., Shanker, A., 2017. Bortezomib augments lymphocyte stimulatory cytokine signaling in the tumor microenvironment to sustain CD8 + T cell antitumor function. Oncotarget 8, 8604–8621.

Peng, Y.B., Zhao, Z.L., Liu, T., Xie, G.J., Jin, C., Deng, T.G., Sun, Y., Li, X., Hu, X.X., Zhang, X.B., Ye, M., Tan, W.H., 2017. A multi-mitochondrial anticancer agent that selectively kills Cancer cells and overcomes drug resistance. ChemMedChem 12, 250–256.

Pitceathly, R.D., Viscomi, C., 2016. Effects of ketosis in mitochondrial myopathy: potential benefits of a mitotoxic diet. EMBO Mol. Med. 8, 1231–1233.

Raina, K., Agarwal, C., Wadhwa, R., Serkova, N.J., Agarwal, R., 2013. Energy deprivation by silibinin in colorectal cancer cells: a double-edged sword targeting both apoptotic and autophagic machineries. Autophagy 9, 697–713.

Ranjan, K., Pathak, C., 2016. Expression of FADD and cFLIPL balances mitochondrial integrity and redox signaling to substantiate apoptotic cell death. Mol. Cell. Biochem. 422, 135–150.
Roshan, M., Kabekkodu, S.P., Vijaya, P.H., Manjunath, K., Graw, J., Gopinath, P.M., Satyamoorthy, K., 2012. Analysis of mitochondrial DNA variations in Indian patients with congenital cataract. Mol. Vis. 18, 181–193.
Saha, S., Panigrahi, D.P., Patil, S., Bhutia, S.K., 2018. Autophagy in health and disease: a comprehensive review. Biomed Pharmacother 104, 485–495.
Saini, S.K., Kalaiarasan, P., Singh, R.K., Manvati, S., Bamezai, R.N.K., 2018. MicroRNA (hsa-miR-19b-2-5p) targets key mitochondrial biogenesis genes-a bioinformatics analysis. Mitochondrion. http://dx.doi.org/10.1016/j.mito.2018.04.005. (2018 Apr 17, pii: S1567-7249(18)30019-9).
Samanta, A., Kumar, P., Machhua, S., Rao, G.N., Pal, A., 2014. Incidence of cystoid macular oedema in diabetic patients after phacoemulsification and free radical link to its pathogenesis. Br. J. Ophthalmol. 98, 1266–1272.
Schopf, B., Schafer, G., Weber, A., Talasz, H., Eder, I.E., Klocker, H., Gnaiger, E., 2016. Oxidative phosphorylation and mitochondrial function differ between human prostate tissue and cultured cells. FEBS J. 283, 2181–2196.
Singh, K.K., Modica-Napolitano, J.S., 2017. Special Issue: Mitochondria in Cancer. Semin. Cancer Biol. 47, iv–vi.
Singh, R.P., Thangaraj, K., Kale, R.K., 2014. Mitochondria in health and disease. Mitochondrion 16, 1.
Sonam, K., Bindu, P.S., Srinivas Bharath, M.M., Govindaraj, P., Gayathri, N., Arvinda, H.R., Chiplunkar, S., Nagappa, M., Sinha, S., Khan, N.A., Nunia, V., Paramasivam, A., Thangaraj, K., Taly, A.B., 2017. Mitochondrial oxidative phosphorylation disorders in children: phenotypic, genotypic and biochemical correlations in 85 patients from South India. Mitochondrion 32, 42–49.
Vandana, V.P., Bindu, P.S., Sonam, K., Govindaraj, P., Taly, A.B., Gayathri, N., Chiplunkar, S., Govindaraju, C., Arvinda, H.R., Nagappa, M., Sinha, S., Thangaraj, K., 2016. Audiological manifestations in mitochondrial encephalomyopathy lactic acidosis and stroke like episodes (MELAS) syndrome. Clin. Neurol. Neurosurg. 148, 17–21.
Vanniarajan, A., Govindaraj, P., Carlus, S.J., Aruna, M., Aruna, P., Kumar, A., Jayakar, R.I., Lionel, A.C., Gupta, S., Rao, L., Gupta, N.J., Chakravarthy, B., Deenadayal, M., Selvaraj, K., Andal, S., Reddy, B.M., Singh, L., Thangaraj, K., 2011. Mitochondrial DNA variations associated with recurrent pregnancy loss among Indian women. Mitochondrion 11, 450–456.
Waseem, M., Tabassum, H., Parvez, S., 2016. Melatonin modulates permeability transition pore and 5-hydroxydecanoate induced KATP channel inhibition in isolated brain mitochondria. Mitochondrion 31, 1–8.
Zhang, H., Singh, K.K., 2014. Global genetic determinants of mitochondrial DNA copy number. PLoS One 9, e105242.
Zi Xu, Y.X., Ande, S.R., Mishra, S., 2018. Prohibitin: a new player in immunometabolism and in linking obesity and inflammation with cancer. Cancer Lett. 415, 208–216.